

Preparation and Properties of Polyimide-Polysiloxane Hybrids Using Sol-Gel Method

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Polyimide (PI) was successfully hybridized with polysiloxanes by synchronizing two reactions; imidization of poly(amide acid) (PAA) and sol-gel process of diethoxysilanes. Diethoxydimethylsilane, diethoxymethylphenylsilane, and diethoxydiphenylsilane were used as precursors for the formation of polydimethylsiloxane (PDMS), polymethylphenylsiloxane (PMPS), and polydiphenylsiloxane (PDPS), respectively. The progress of sol-gel reaction was confirmed by IR. The compatibility of polysiloxanes with polyimide was studied by scanning electron microscopy, which revealed better compatibility of phenyl group containing polysiloxanes. The effect of the pendant group of polysiloxane on the mechanical and thermal properties of PI-polysiloxane hybrids was studied. Tensile strength of PI were increased by the addition of small amount of polysiloxanes. Hybrid containing 3 wt% of PDMS gave the highest tensile properties. Dynamic viscoelastic analysis of the PI-polysiloxane hybrid revealed two glass transition temperatures corresponding to polysiloxane and PI components. Thermogravimetric analysis revealed higher thermal stability by the hybridization.

Keywords: polyimide / sol-gel / hybrid / polysiloxane / thermal property

1. Introduction

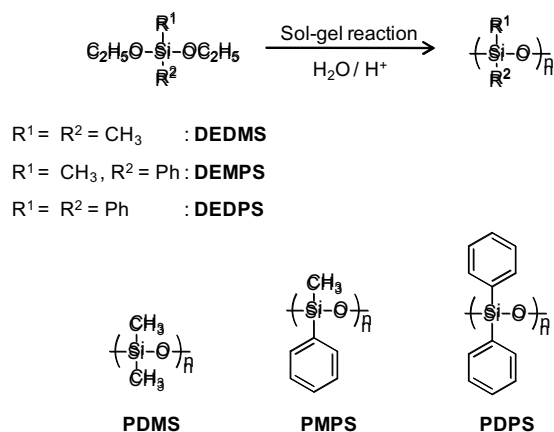
Polyimide (PI) is one of the most important super-engineering plastic because of the outstanding properties, such as excellent thermal and mechanical properties, good chemical resistance, and low dielectric constant. Now, PI is extensively used in the field of aerospace, electronic, and other high-tech industries [1, 2]. Research interests in PIs have been growing in response to the increase of technology-focused applications of PIs. In the past decade, significant efforts have focused on synthesizing PIs with further enhanced thermal and mechanical properties [3]. The performance of PIs can be improved by designing novel polyimides, polymer alloys, and the organic-inorganic hybrid. Concerning the organic-inorganic hybrid using PI, various types of inorganic component such as clay [4], silica [5], metal oxide [6], and metal nanoparticles [7] have been used.

Polydimethylsiloxane (PDMS), an important inorganic polymer, shows high flexibility and good thermal stability [8]. The introduction of PDMS into polymeric materials is expected to improve the

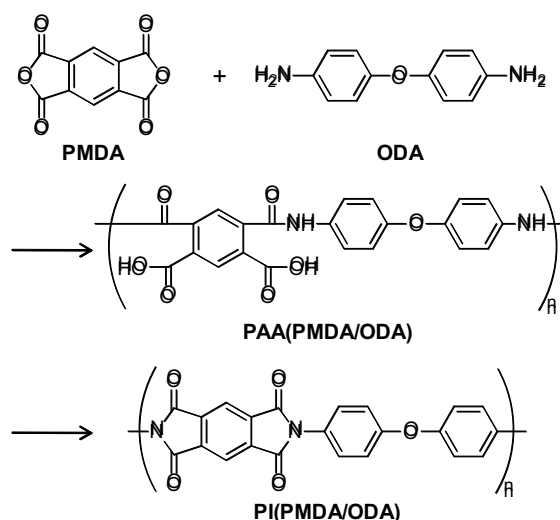
performance of the polymer. However, PDMS shows low dispersability in an organic polymer matrix because of its incompatibility with organic polymers [9]. One approach to improve the compatibility and dispersability of PDMS is copolymerization. A successful example is the formation of transparent poly(imide-siloxane) copolymers [10]. However, heat resistance of poly(imide-siloxane) copolymers is low. This is because the siloxane diamine contains trimethylene units, and reduces the heat resistance. Previously, by utilizing sol-gel reaction for the formation of PDMS, we successfully prepared PI-PDMS hybrids which revealed enhancement of not only toughness but also thermal properties with inclusion of small amount of PDMS [11]. However, further increase of the PDMS content decreased tensile properties probably because of the aggregation of PDMS. If aggregation can be avoided, further increase of the polysiloxane content becomes possible, and the properties of the hybrids should be enhanced.

Polysiloxane having phenyl group as the pendant is expected to enhance the compatibility

between PI and polysiloxane because of the π - π interaction between the phenyl groups of the two components. As a result, higher amount of polysiloxane can be introduced without aggregation in the hybrids, and improved properties are expected for the hybrids. In this paper, we report a series of PI-polysiloxane hybrids using different types of polysiloxanes containing phenyl groups such as polymethylphenylsiloxane (PMPS) and polydiphenylsiloxane (PDPS), and compared the properties of the hybrids with those of the PI-PDMS hybrid. The polysiloxanes were prepared by sol-gel reaction (Scheme 1). PI was prepared from pyromellitic dianhydride (PMDA) and 4,4'-oxydianiline (ODA) (Scheme 2).



Scheme 1. Synthesis of polysiloxanes by sol-gel reaction.



Scheme 2. Synthesis of PI(PMDA/ODA).

2. Experimental

2.1. Reagents

PMDA and ODA obtained from Tokyo Kasei Kogyo (Tokyo, Japan) were purified by sublimation just before use. 1-Methyl-2-pyrrolidinone (NMP) obtained from Wako Pure Chemical Industries (Osaka, Japan) was distilled under reduced nitrogen pressure over CaH_2 just before use. Diethoxydimethylsilane (DEDMS), diethoxymethylphenylsilane (DEMPS), and diethoxydiphenylsilane (DEDPS) were obtained from Wako Pure Chemical Industries (Osaka, Japan), and were used as received.

2.2. Preparation of poly(amide acid)

Poly(amide acid) (PAA) was prepared from PMDA and ODA using NMP as solvent [11]. The inherent viscosity of the PAA solution was in the range of 1.0~ 1.2 dl/g (0.5 g/dl in NMP at 30 °C).

2.3. Preparation of PI-polysiloxane hybrids

As a typical experimental procedure for the preparation of PI-polysiloxane hybrid, hybrid containing 5 wt% of PMPS (PI-PMPS (5%)) is described below.

Into the PAA (0.99 g, 2.4 unit mmol) solution in NMP (9 ml) were added DEMPS (77.8 mg, 0.37 mmol) and water (13.2 mg, 0.74 mmol). The mixture was stirred for 24 h at room temperature to produce a transparent yellow solution that was then cast onto a glass plate, dried at 60 °C for 16 h under vacuum, cured at 100 and 200 °C for 1 h each in a vacuum oven, and then cured at 300 and 350 °C for 1 h each in an air oven. PI-PMPS (5%) hybrid was obtained as sunny yellow transparent film. Other PI-polysiloxane hybrids were prepared similarly.

2.4. Measurements

Viscosity of PAA was measured using Ubbelohde viscometer at a concentration of 0.5 g/dl in NMP at 30 °C. IR spectra were obtained using a Jasco FT/IR-420 spectrometer. UV-Vis spectra were measured on a JASCO V-550 spectrophotometer. Scanning electron microscopy (SEM) studies were performed using a Hitachi S-4800 scanning electron microscope with an accelerating voltage of 3 kV and emission current of 10 μA . The fracture surfaces of the PI hybrids were fixed by Ag paste and were sputter-coated with Pt-Pd under an electric current of 20 mA at 6 Pa for 60 s. Tensile properties were recorded with an Imada Seisaku-sho Model SV-3 at a crosshead speed of 1 mm/min using films about 2 cm long.

The tensile properties of each sample were determined from an average of at least 10 tests. Dynamic viscoelastic measurements (DMA) were conducted on an Orientec Automatic Dynamic Viscoelastomer Rheovibron model DDV-01FP at 35 Hz and at a heating rate of 4 °C/min. Thermogravimetric analysis (TGA) was performed with a Rigaku Thermo Plus 2TG-DTA TG8120 at a heating rate of 5 °C/min under argon atmosphere.

3. Results and Discussion

3.1. Preparation of PI-polysiloxane hybrids

PI-polysiloxane hybrids were prepared by combining the sol-gel reaction of diethoxysilanes and imidization of PAA. First, PAA, diethoxysilane, and water were dissolved in NMP at room temperature, and then the cast films were thermally treated up to 350 °C for imidization of PAA and the sol-gel reaction of diethoxysilane. In this case, carboxylic group of PAA acts as acid catalyst, and further addition of acid catalyst is not needed [12]. The feed of PAA and diethoxysilane was determined by assuming complete conversion of PAA into PI and diethoxysilanes into polysiloxanes. The sample code is abbreviated based on the feed of the polysiloxane in the hybrids. For example, PI-PDMS (10%) stands for PI-PDMS hybrid with

10 wt% of PDMS feed. Table 1 shows the formulation for the preparation of PI-polysiloxane hybrids.

Figure 1 shows IR spectra of the hybrids at 10 wt% of polysiloxane feed after various cure stages. The absorptions corresponding to PAA appeared at 1660 (amide carbonyl) and 1542 cm⁻¹ (C–NH stretching) after drying at 60 °C. These absorptions decreased and almost disappeared after curing at 200 °C (Figure 1a~ 1c). Instead, the appearance of the characteristic absorptions of the imide groups at 1776 (C=O symmetric stretching), 1720 (C=O asymmetric stretching), and 1378 cm⁻¹ (C–N stretching in imide ring) after the 200 °C cure indicates the progress of imidization by the curing (Figure 1a~ 1c). The introduction of PDMS into the hybrid was confirmed by the absorptions at 805 (Si–CH₃ stretching) and 1016 cm⁻¹ (Si–O–Si stretching) (Figure 1a). The introduction of PMPS was confirmed by the absorptions at 697 (Si-phenyl stretching), 805 (Si–CH₃ stretching), and 1016 cm⁻¹ (Si–O–Si stretching) (Figure 1b). The introduction of PDPS was confirmed by the absorptions at 697 (Si-phenyl stretching) and 1016 cm⁻¹ (Si–O–Si stretching) (Figure 1c).

Table 1. Formulation for the preparation of PI-Polysiloxane hybrids.

Run	Code	Theoretical Polysiloxane content / wt%	PI / g (mmol)	Diethoxysilane / g (mmol)	H ₂ O / mg (mmol)
1	PI	-	1.05 (2.50)	-	-
2	PI-PDMS (1%)	1	1.04 (2.47)	0.015 (0.10)	3.60 (0.20)
3	PI-PDMS (3%)	3	1.02 (2.42)	0.045 (0.31)	11.16 (0.62)
4	PI-PDMS (5%)	5	0.99 (2.37)	0.076 (0.51)	18.36 (1.02)
5	PI-PDMS (10%)	10	0.94 (2.25)	0.151 (1.02)	36.72 (2.04)
6	PI-PMPS (1%)	1	1.04 (2.47)	0.015 (0.07)	2.64 (0.14)
7	PI-PMPS (3%)	3	1.02 (2.42)	0.046 (0.22)	7.93 (0.44)
8	PI-PMPS (5%)	5	0.99 (2.37)	0.078 (0.37)	13.20 (0.74)
9	PI-PMPS (10%)	10	0.94 (2.25)	0.154 (0.73)	26.42 (1.46)
10	PI-PDPS (1%)	1	1.04 (2.47)	0.013 (0.05)	0.91 (0.10)
11	PI-PDPS (3%)	3	1.02 (2.42)	0.040 (0.15)	2.73 (0.30)
12	PI-PDPS (5%)	5	0.99 (2.37)	0.069 (0.25)	4.54 (0.50)
13	PI-PDPS (10%)	10	0.94 (2.25)	0.137 (0.50)	9.07 (1.00)

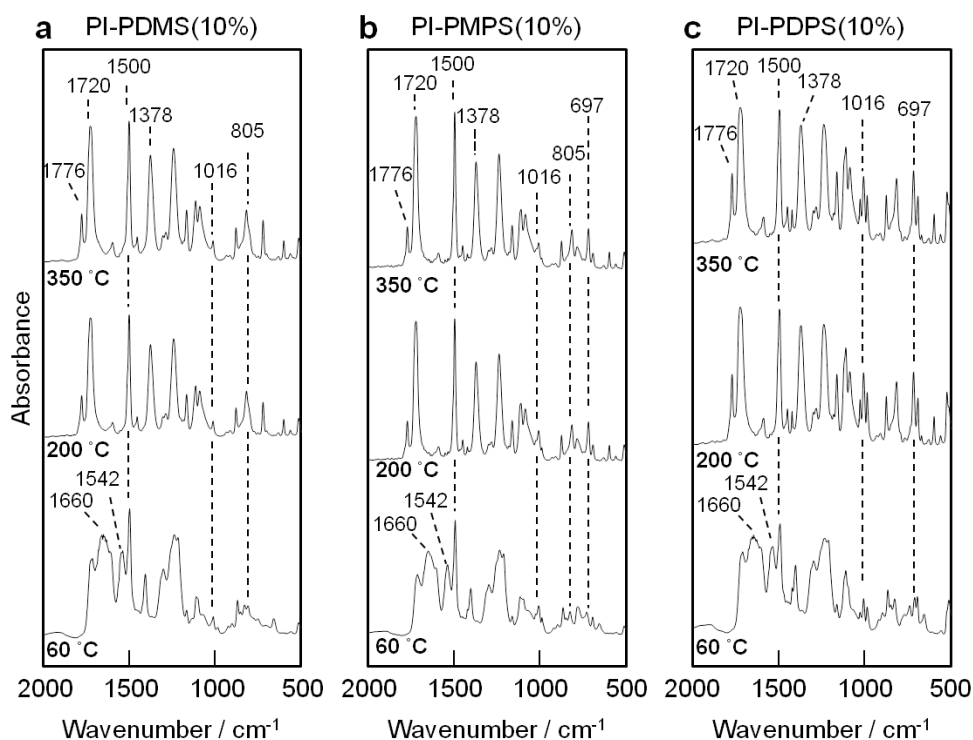


Figure 1. IR spectra of PI-PDMS(10%) (a), PI-PMPS(10%) (b) and PI-PDPS(10%) (c) hybrids after curing at 60, 200, and 350 °C.

The progress of imidization after each curing stage was followed by the increase of the ratio of the imide absorbance at 1776 cm^{-1} against the absorbance of aromatics at 1500 cm^{-1} , and is shown in Figure 2A. It is clearly shown that imidization proceeded by the cure, and was almost completed after curing at 200 °C . The progress of imidization was almost the same without depending on the type of polysiloxanes. Figure 2B shows the change of the ratio of the absorbance at 1016 cm^{-1} with curing

temperature. The Si–O–Si absorbance increased with the progress of cure, suggesting that the condensation of diethoxysilanes occurred during the cure, possibly increasing the molecular weight of polysiloxanes [13, 14]. All the polysiloxanes showed similar trend. Therefore, Figure 2 shows that both reactions, imidization and the sol-gel reaction, proceeded in a similar temperature range during the cure. It is probably due to this synchronized progress of the two reactions that we

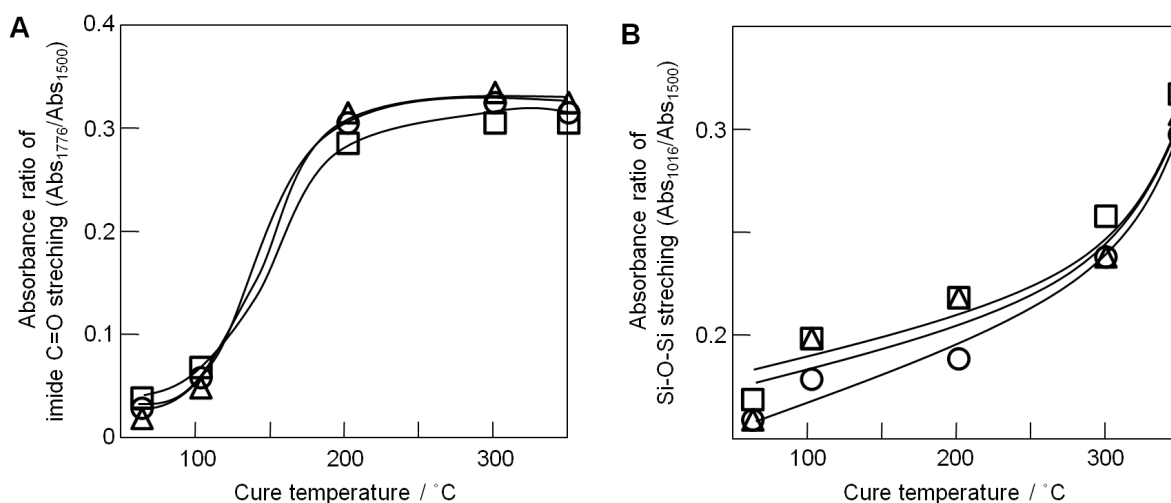


Figure 2. Progress of imidization of PAA and sol-gel reaction of diethoxysilanes estimated from IR spectra by the absorbance of imide C=O stretching (A) and Si-O-Si stretching (B) in PI-PDMS(10%) (○), PI-PMPS(10%) (□), and PI-PDPS(10%) (△) hybrids.

successfully obtained homogeneous hybrid films.

3.2. Effect of the chemical structure of polysiloxanes on the compatibility with PI

Figure 3A shows UV-vis spectra of pristine PI, PI-PDMS (10%), PI-PMPS (10%) and PI-PDPS (10%) films. The transmittance of the hybrid films at 700 nm is shown in Figure 3B. The pristine PI film (ca. 27 μm thickness) showed high transparency and transmitted 85 % of 700 nm light. The transmittance of the PI-PDMS hybrid significantly decreased even as small as 5 wt% of PDMS feed (Figure 3B). PI-PDMS (10%) was transmitting only 1% of 700 nm light. On the other hand, PI-PMPS and PI-PDPS showed much higher transparency. For example, PI-PMPS (10%) and PI-PDPS (10%) were transmitting 70% and 80% of 700 nm light, respectively. The PI-PMPS hybrids showed high transparency up to 20 % feed, and the PI-PDPS hybrid showed high optical transmittance even at the polysiloxane content as high as 50 % feed (Figure 3B). The higher transparency of the PI-polysiloxane hybrids was achieved by the polysiloxane with higher aromatic content. This is probably because of the enhanced compatibility due to the phenyl group in PMPS and PDPS.

The transparency of the hybrid films comes from the dispersion of polysiloxanes in the PI matrix. Thus, we investigated the fracture surface morphology of PI-polysiloxane hybrids by SEM (Figure 4). The fracture surface of pristine PI was smooth. The PI-PDMS hybrid films revealed two phases, because of the low compatibility of PDMS with PI even at 3 wt% content (Figure 4a). The

diameter of the dispersed spherical PDMS domains was 0.2~0.5 μm , relating to the opaque appearance of the film, and indicating micrometer-scale phase separation (Figure 4a~c). On the other hand, the PI-PMPS and PI-PDPS hybrids revealed homogeneous phase without phase separation up to 10 and 20 wt% feed, respectively, indicating the high compatibility of the polysiloxanes with PI, probably because of the π - π interaction between the phenyl groups in the polysiloxanes (Figure 4d~g).

3.3. Thermal and mechanical properties of the PI-polysiloxane hybrids

Viscoelastic properties of the PI-polysiloxane hybrids were examined by DMA. DMA of hybrids containing 10% polysiloxanes are shown in Figure 5. Storage modulus (E') at room temperature and T_g s are summarized in Table 2. PI-PDMS hybrid has two T_g s [11]. Lower T_g at around -100 $^{\circ}\text{C}$ is derived from PDMS, and higher T_g is derived from PI. The finding of T_g based on PDMS agrees with the SEM observation. Also, we notice slight decrease of E' at the T_g of PDMS. It is also worth noting that, in spite of the addition of PDMS, the higher T_g corresponding to the PI component was maintained above 400 $^{\circ}\text{C}$ due to the micro-phase separation. Lower T_g s derived from PMPS and PDPS were observed around -30 $^{\circ}\text{C}$ and 60 $^{\circ}\text{C}$, respectively, even though SEM did not show domains of the polysiloxanes. Therefore, PMPS and PDPS is considered to form very small domains that was not detected by the SEM measurement. This means all the hybrids were

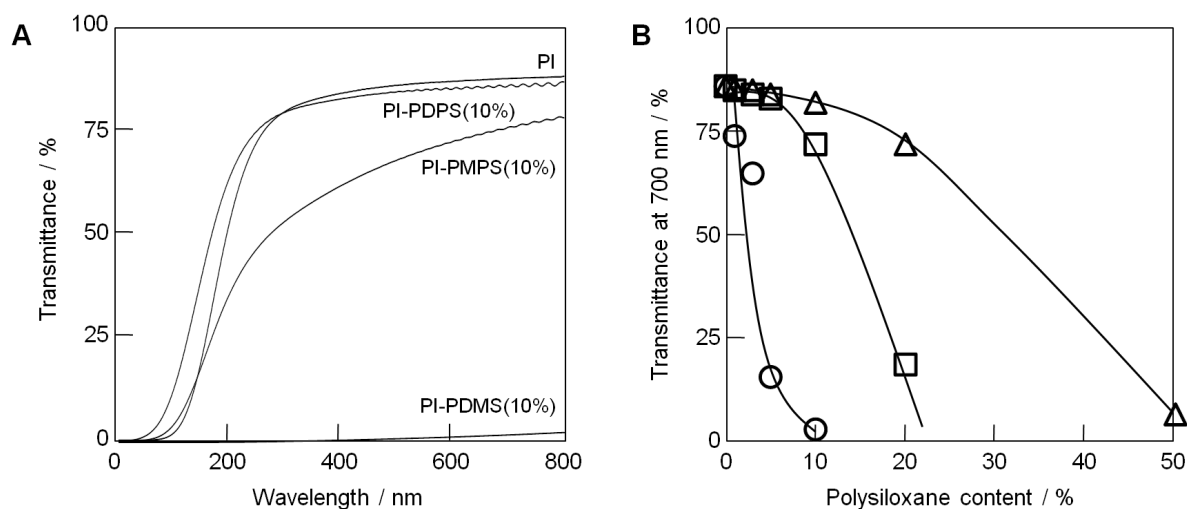


Figure 3. (A) UV-vis spectra of pristine PI, PI-PDMS(10%), PI-PMPS(10%) and PI-PDPS(10%) films. (B) Plots of transmittance at 700 nm of the PI-polysiloxanes hybrids against the polysiloxanes contents for PI-PDMS hybrid (○), PI-PMPS hybrid (□), and PI-PDPS hybrid (△).

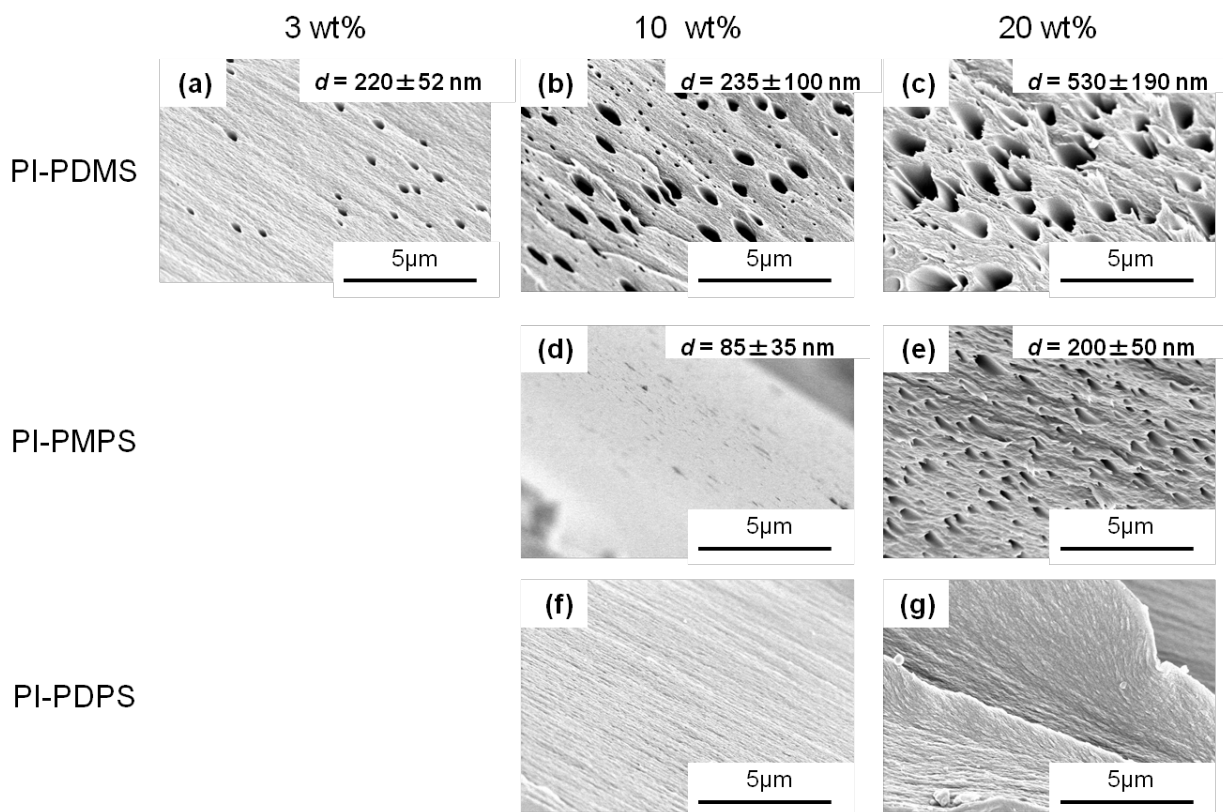


Figure 4. SEM images for the fracture surfaces of PI-PDMS(3%) (a), PI-PDMS(10%) (b), PI-PDMS(20%) (c), PI-PMPS(10%) (d), PI-PMPS(20%) (e), PI-PDPS(10%) (f) and PI-PDPS(20%) (g) hybrids.

phase separated.

Thermal stability of the PI-polysiloxane hybrids was investigated by TGA under argon atmosphere (Figure 6 and Table 2). All the hybrids showed higher degradation temperature and char yields with increasing polysiloxane content, because the polysiloxanes possess high thermal stability (Table 2). The 5% weight loss temperature (T_5) was not much different among the hybrids. The T_{10} s and weight residues at 850 °C of PI-PDPS hybrids are higher than those of PI-PDMS and PI-PMPS hybrids.

Tensile properties of the hybrid films were examined. Figure 7 shows the tensile tests of the PI-polysiloxane hybrids in the form of stress-strain curves. The tensile modulus, tensile strength, and elongation at break against the polysiloxanes content are plotted in Figure 8A, 8B, and 8C, respectively. The pristine PI showed tensile modulus of 2.5 GPa, tensile strength of 160 MPa, and elongation at break of 85 %. By introducing small amount of PDMS component, tensile properties were improved. As a result, the PI-PDMS (3%) film showed highest tensile properties: tensile modulus of 2.8 GPa, tensile strength of 177 MPa, and 96 % elongation at break

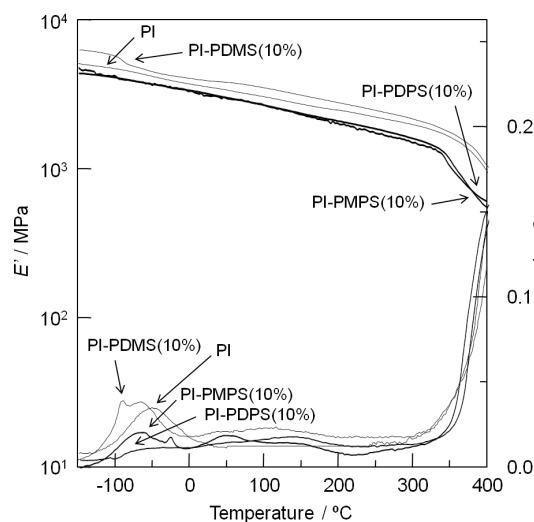


Figure 5. DMA results of pristine PI, PI-PDMS(10%), PI-PMPS(10%) and PI-PDPS(10%) hybrids.

[11]. Interestingly, in spite of the addition of flexible PDMS, the tensile modulus also increased. This suggests that small amount of PDMS promotes the in-plane orientation of PI in the hybrids. A further increase of the PDMS content (5%) started to decrease the tensile properties, probably due to the aggregation of PDMS.

Table 2. Thermal properties of PI-Polysiloxane hybrids.

Code	Theoretical Polysiloxane content / wt%	DMA			TGA		
		E' at R.T. (GPa)	Lower T_g (°C)	Higher T_g (°C)	T_5 (°C)	T_{10} (°C)	Weight residue at 850 °C (%)
PI	-	2.5	-	>400	501	540	52
PI-PDMS (1%)	1	2.6	-	>400	525	555	53
PI-PDMS (3%)	3	2.8	-87	>400	528	559	55
PI-PDMS (5%)	5	2.5	-90	>400	532	559	55
PI-PDMS (10%)	10	2.0	-104	>400	545	561	58
PI-PMPS (1%)	1	2.6	-	>400	528	556	54
PI-PMPS (5%)	5	2.5	-32	>400	530	558	55
PI-PMPS (10%)	10	1.8	-26	>400	543	560	58
PI-PDPS (1%)	1	2.5	-	>400	528	559	54
PI-PDPS (5%)	5	2.3	90	>400	539	558	58
PI-PDPS (10%)	10	2.1	60	>400	543	566	60

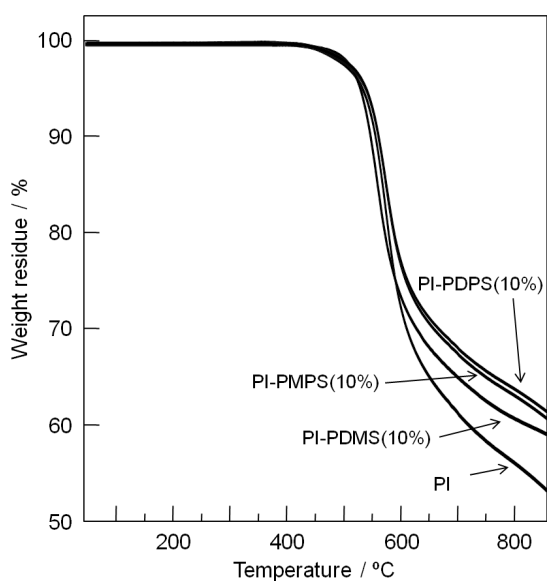


Figure 6. TGA curves of pristine PI, PI-PDMS(10%), PI-PMPS(10%) and PI-PDPS(10%) hybrids.

For PI-PMPS hybrids, PI-PMPS (1%) film showed highest tensile properties: tensile modulus of 2.6 GPa, tensile strength of 170 MPa, and 89 % elongation at break. A further increase of the PMPS content (3%) started to decrease the tensile properties. For PI-PDPS hybrids, PI-PDPS (1%) film showed highest tensile properties: tensile

modulus of 2.4 GPa, tensile strength of 162 MPa, and 80 % elongation at break. A further increase in the PDPS content (3%) started to decrease the tensile properties. Figure 8A shows modulus increased with small amount of PDMS and PMPS, but decreased for all the PI-PDPS hybrids. Figure 8B and 8C indicate the same tendency for tensile strength and elongation at break. PMPS and PDPS were less effective than PDMS for the performance improvement of the hybrids, and PI-PDMS hybrids showed the highest tensile properties.

4. Conclusions

PI-polysiloxane hybrids with different types of polysiloxanes were successfully prepared by synchronizing the imidization of PAA and the sol-gel reaction of diethoxysilanes. The phenyl group of polysiloxanes enhanced their compatibility with PI and transparency of the hybrid films. The hybrid films with small amount of PDMS enhanced the tensile properties, and introduction of 3 wt% PDMS showed highest tensile properties among the hybrids. Furthermore, the degradation temperature and weight residue at 850 °C also increased by the hybridization. The PI-PDPS hybrid showed most effective enhancement of thermal stability. The pendant

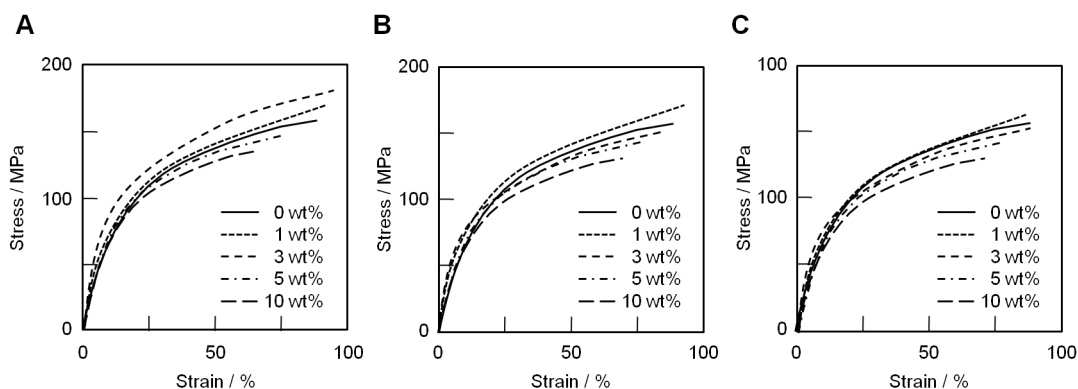


Figure 7. Stress-strain curves of PI-PDMS hybrids (A), PI-PMPS hybrids (B), and PI-PDPS hybrids (C).

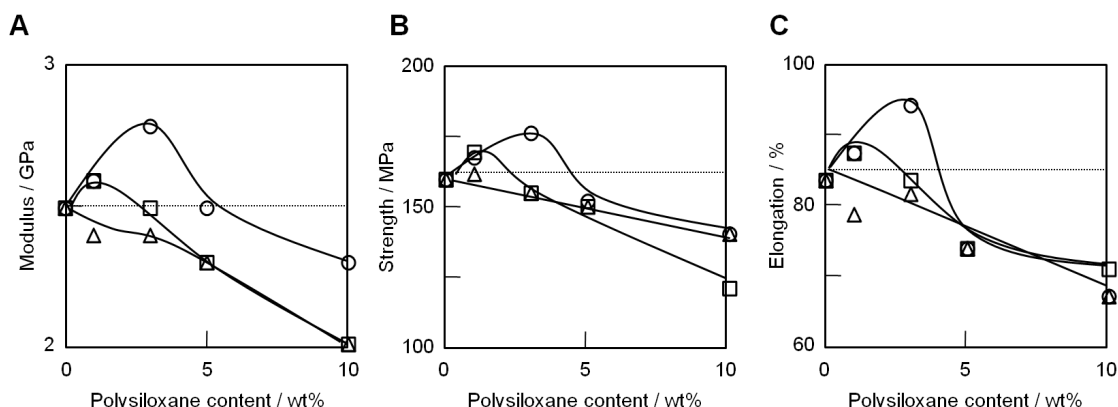


Figure 8. Plots of tensile modulus (A), tensile strength (B), and elongation at break (C) of the PI-polysiloxane hybrids against the polysiloxane content. (○): PI-PDMS hybrid, (□): PI-PMPS hybrid and (△): PI-PDPS hybrid.

group of the polysiloxane strongly affects the optical, thermal, and mechanical properties, and those properties of the hybrids can be controlled by the type of polysiloxanes.

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